

Statistics 240 Homework Simulation: Introduction

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Instructions

1. Fill in this `.qmd` file by modifying the given code, removing *italicized* instructions, and narrating your solutions.
2. Render this `.qmd` file to a `.html` file in RStudio.
3. Open the `.html` file with a web browser and save as a `.pdf` by going to the print menu and selecting "save as PDF."
4. Submit the `.pdf` file as with other homework assignments.

Exercise 1

I have two fair, six-sided dice: a red die and a green die. If both dice are thrown, what is the probability that the red die will show a higher number than the green die?

a) Calculate this probability by hand.

Given that the green shows up as 1, red can be 2,3,4,5,6 – $(1/6) \cdot (5/6)$

Given that the green shows up as 2, red can be 3,4,5,6 – $(1/6) \cdot (4/6)$

...

Given that the green shows up as 5, red can be 6 $(1/6) \cdot (1/6)$

Given that the green shows up as 6, there is no chance for red to be bigger $(1/6) \cdot (0/6)$

Summing these probabilities will give the total: $(5+4+3+2+1+0)/36 = 15/36 = .4167$

b) Write an R function that simulates rolling the two dice and reports whether the red die shows a higher number than the green die.

Here is code for a function to simulate rolling a single six-sided die.

```
roll <- function() {  
  outcomes <- 1:6  
  probs <- rep(1.0/6.0, 6)  
  sample(outcomes, size = 1, prob = probs)  
}
```

```
roll()
```

```
[1] 5
```

Now, create a function that uses `roll()` to simulate an outcome for each of the red and green dice and reports whether the red is higher than the green.

```
red_higher <- function() {
  red_roll <- roll() # generate red dice value
  green_roll <- roll() # generate green dice value
  return(red_roll > green_roll) # return true if red value is higher than green value
}
```

c) Simulate this experiment 10,000 times and report the proportion of times the red die was higher. Compare the result to the theoretical probability.

Use the `replicate()` function as demonstrated in the slides.

```
samp <- replicate(10000, red_higher()) # store the samples here
mean(samp) # use the samples to calculate the proportion
```

```
[1] 0.4174
```

Exercise 2

An emergency phone line receives calls for both fire and non-fire incidents. Fire-related calls occur 30% of the time. Each phone call is independent of the other phone calls.

a) Write an R function that accepts argument `n`, simulates `n` independent phone calls, and reports whether each call is related to a fire incident.

Complete the code below to simulate `n` calls and return the sequence of outcomes.

```
calls <- function(n) {
  sample(c("fire", "other"), size = n, replace = TRUE, prob = c(.3, .7))
}
```

b) Repeatedly simulate sequences of two calls to answer the following question: What is the probability that among the next two calls, exactly one is fire related?

First, you will need a function that accepts `n`, runs `calls(n)`, and counts how many of the simulated calls are fire related. Then, use the `replicate()` function.

```
count_calls <- function(n) {
  is_fire <- calls(n) == "fire"
  sum(is_fire) # now count how many elements of is_fire are TRUE
}

call_list <- replicate(10000, count_calls(2)) # replicate count_calls many times and store samples

exact1 <- call_list == 1 # use the samples to calculate the proportion
mean(exact1)
```

[1] 0.4162

c) Repeatedly simulate sequences of five calls to answer the following question: What is the probability that among the next five calls, at least three are fire related?

```
call_list <- replicate(10000, count_calls(5)) # replicate count_calls (with five calls) many times
morethan3 <- call_list >= 3
mean(morethan3) # use the samples to calculate the proportion
```

[1] 0.1578